Electricity Bill Calculator

B.Tech Python Programming – Activity-Based Learning

Python source code only. Save or copy the program into a .py file and run it with Python 3.

```
while True:
    try:
        units = float(input("Enter electricity units consumed: "))
        if units >= 0:
            break
        print("Units cannot be negative.")
    except ValueError:
        print("Enter a valid number.")

if units <= 100:
    energy_charge = units * 1.50
elif units <= 200:
    energy_charge = 100 * 1.50 + (units - 100) * 2.50
elif units <= 300:
    energy_charge = 100 * 1.50 + 100 * 2.50 + (units - 200) * 4.00
else:
    energy_charge = (100 * 1.50 + 100 * 2.50
                     + 100 * 4.00 + (units - 300) * 6.00)

fixed_charge = 50
total_bill = energy_charge + fixed_charge

print("\n----- ELECTRICITY BILL -----")
print(f"Units consumed: {units:g}")
print(f"Energy charge: Rs. {energy_charge:.2f}")
print(f"Fixed charge: Rs. {fixed_charge:.2f}")
print(f"Total bill: Rs. {total_bill:.2f}")
```